Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/04/2017
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	Constrained Vapor Bubble Flight Experiment II – Binary Mixtures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	plawsky@rpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6049
Organization Name:	Rensselaer Polytechnic Institute		
PI Address 1:	Department of Chemical and Biological Engineering		
PI Address 2:	110 Eighth Street, RI-122		
PI Web Page:	http://www.rpi.edu/		
City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2013	**End Date:**	03/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 3/31/2019 per NSSC information (Ed., 12/7/17)		
Key Personnel Changes/Previous PI:	July 2017 report: Graduate student Akshay Kundan graduated; Graduate student Thao Nguyen graduated; Graduate student Jiaheng Yu joined the group.		
COI Name (Institution):	Wayner, Peter Ph.D. (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX13AQ78G		
Performance Goal No.:			
Performance Goal Text:			
	The thermophysical principles underlying change-of-phase heat transfer systems controlled by interfacial phenomena under microgravity conditions are not well understood. As a result, related passive engineering systems like heat pipes have not been optimized and it is unclear whether recent advances in the design of these systems based on ground-based research using fluid mixtures can be translated to microgravity systems. Herein, we propose basic experimental and theoretical studies of the nonisothermal constrained vapor bubble (CVB-2) using a mixture of pentane and isohexane under microgravity conditions to help remedy this undesirable situation. This proposed study is multi-faceted: 1) it is a basic scientific study in interfacial phenomena, microgravity fluid physics, and thermodynamics; 2) it is a basic study in thermal and mass transport; and 3) it is a study of a passive heat exchanger. Although the basic scientific Facets (1) and (2) are emphasized, the research naturally leads to the development of a passive heat exchanger optimized for		

Task Description:	microgravity conditions. The proposed optical study of a vapor bubble constrained in a transparent glass cell under nonisothermal conditions will increase the basic understanding of transport systems controlled by interfacial phenomena. The pressure gradient field will be obtained optically whereas the temperature field will be obtained using thermal sensors. Due to the sensitivity of systems of the proposed relatively large size to gravity, to small temperature and pressure gradients, and to changes in fluid composition, these transport systems need to be studied under microgravity conditions. An "axisymmetric" system with a small Bond number is required. Direct extensions to the microgravity environment first investigated in CVB-1 are proposed. The augmented Young-Laplace equation and the Kelvin-Clapeyron model will be used to help understand and quantify how the system operates. The immediate macroscopic objectives of CVB-2 are to determine the stability, the fluid flow characteristics, the average heat transfer coefficient in the evaporator and condenser sections, and the heat conductance of the CVB as a function of the heat input using the new fluid mixture. The immediate microscopic objective is to determine the detailed characteristics of the transport processes in the curved liquid film. The local conditions under which cavitation and instability occur with the formation of a dry region will be determined as a function of heat flux, film thickness, and stress. High fluxes and relatively large systems with small capillary pressures will be emphasized. The data using the pentane/isohexane mixture will be compared with the pure pentane data obtained in CVB-1 and ground-based data on identical fluid mixtures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work addresses fundamental questions regarding phase-change heat and mass transfer operations that are becoming increasingly important in a wide variety of fields ranging from energy conversion, to microelectronics cooling, micro-electromechanical systems (MEMS) devices, self-assembly, and space flight. Of particular concern to NASA are microchannel and heat pipe systems used for thermal management and perhaps separation processes. Such systems are attractive for long-term space flights because they generally require no moving parts and are potentially maintenance free. The ability to control transport processes in an extended meniscus will be of great utility both on Earth and in space, whether we are looking for enhanced heat transfer, coating, or self-assembly applications. Some of our preliminary ground-based experiments concerning the microscopic evaporating meniscus, performed as part of the existing CVB-1 flight experiment program, have indicated that using a fluid mixture, with the minority component being the less volatile, higher surface tension liquid, offers the opportunity to control fluid flow and heat transfer in an evaporating meniscus and thus enhance the performance of heat pipes. The surface concentration gradient will also enhance axial fluid flow in the corners. The extension of the CVB-1 experiment to mixtures allows us to isolate the Marangoni effects from the effects of gravity and begin to understand how to exploit Marangoni flows for improving the performance of heat pipes in space and on Earth.
Task Progress:	SUMMARY During the course of this year we have been vigorously working on analyzing the data from both CVB-2 and CVB-1 flight experiments. We have focused most recently on 50x data to see what we could recover from those sets of experiments and to see what differences we could see at that magnification level between CVB-1 and 2. We have also expanded our focus to include how the cooler temperature affects performance and have developed numerical models for meniscus oscillation. We have discovered the reason behind liquid accumulation at the heater end of the CVB and traced it to instabilities in the thin liquid film that promote condensation even at very high temperatures. Our numerical oscillation model reproduces the gross features of what we observe experimentally and curiously points to the prospect that to achieve maximum heat transfer via evaporation, we must also have condensation. We have been continued to write papers based on CVB1 and CVB2 data, and continue to present the information at as many conferences as we could reasonably attend. YEARLY HIGHLIGHTS 1. We spent a considerable amount of time regressing and analyzing the data from the high heat input experiments run aboard the International Space Station (ISS) in late 2010 using the 30 and 40 mm module CVB1 experiments on the ISS and developing enhanced thermal models to determine the internal heat transfer rates within the cuvette. This has resulted in a publication in the Journal of Colloid and Interface Science. 2. We have analyzed the low resolution (surveillance and 10x) images associated with CVB-2 and have developed mathematical models tying the images and the temperature profile to the fluid composition and the performance of the evaporator section of the heat pipe. This has resulted in a publication in the International Journal of Heat and Mass Transfer. 3. We have recently uncovered similar data from CVB-1. This is currently being analyzed but shows that the cooler temperature even for the pure fluid, exerts a tremendous influence on the processes occurring at the heater end. We are still analyzing the data and will be submitting a paper on this topic in 2017. 4. The high heat input data on CVB-1 has led us to discover new phenomena. The first is wholesale condensation at the heater end of the device which we believe serves as the source for fluid accumulation at that end. This resulted in a Physical Review Letters paper in 2015. This year we believe we determined the root cause of the condensation and accumulation. We believe it is due to an instability in the thin liquid film due to disjoining pressure and Marangoni effects. If so, it would validate one of the earliest, scientific justifications for running CVB-1. This work resulted in a second paper in Physical Review Letters in 2017 that was an Editor's Choice article. 4. The high heat data has also led to a self-organizing, reverse jet that forms on the flat surfaces of the CVB (CVB-1) near the heater end. This also drives fluid from regions where evaporation is small (near the heater) to where it is large (near the central drop). A combination of condensation, evaporation, and Marangoni flow we believe leads to the behavior. This topic is the subject of at least one paper that will be submitted in the Fall of 2017. 5. We submitted a proposal to NASA to use the Physical Sciences Informatics database to continue analyzing and modeling results from CVB1. The Principal Investigator on that proposal was Vladimir Ajaev of Southern Methodist University. This proposal was recently funded and should help us with a sound mathematical framework for trying to

describe the phenomena occurring in the Physical Review Letters papers.

6. We submitted a proposal to the National Science Foundation (NSF)/CASIS fluid dynamics call. The goal was to expand on CVB-2 and fly a nearly 50:50 mixture of pentane and isohexane on the ISS. In this region, odd surface tension effects were observed in ground-based experiments, and very turbulent heat transfer performance was also observed. The project has in-kind support from Intel and also Jay Mann and Bill Meyer. The proposal was funded in September 2016.

7. We presented results at 4 meetings/seminars with three more scheduled for the rest of 2017. One is a plenary talk at the 2017 International Conference on Nano, Micro, and Minichannels.

8. Thao Nguyen defended her thesis in the Spring of 2017 and will graduating in the summer of 2017.

9. We have developed a model for the oscillating meniscus. The curious result of that work was that to enhance evaporation, condensation is required. The periodic flooding of the surface promotes an increase in evaporative heat transfer. This will be the subject of a paper Fall 2017 and another issue to study as part of the Physical Sciences Informatics Database grant.

FUTURE WORK

As outlined in the highlights section, we will continue to analyze results from CVB1 and CVB2. We are preparing for CVB3 funded by NSF/CASIS and are working on ground-based systems to do preliminary experimentation and develop the experimental matrix. We recently receive a grant to use the Physical Sciences Informatics Database to develop numerical models for the phenomena we observed in CVB1. We will be working to extend our current models to fulfill that task. Our work will also be focused on determining practical applications from CVB 1& 2 and how to design new, wickless heat pipes for space and terrestrial applications.

The work we have conducted on CVB1 & 2 has far exceeded the original goals for the project and continues to offer up new and intriguing observations to explain. We will be writing for a one-year no-cost extension to finalize the CVB2 project.

Bibliography Type:	Description: (Last Updated: 10/14/2025)
Abstracts for Journals and Proceedings	Nguyen TT, Kundan A, Wayner PC Jr, Plawsky JL. "Experimental study of the contact line region for a pure fluid and binary fluid mixture on the ISS." 2016 AIChE National Meeting, San Francisco, CA, November 13-18, 2016. 2016 AIChE National Meeting, San Francisco, CA, November 13-18, 2016. https://aiche.confex.com/aiche/2016/webprogram/Paper470090.html ; accessed 7/17/2017. , Nov-2016
Abstracts for Journals and Proceedings	Nguyen TT, Kundan A, Wayner PC Jr, Plawsky JL. "Experimental Study of the Contact Line Region for Pure and Binary Fluid Mixture on ISS." 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Nguyen TT, Kundan A, Wayner PC Jr, Plawsky JL, Chao DF, Sicker RJ. "Experimental study of the heated contact line region for a pure fluid and binary fluid mixture in microgravity." J Colloid Interface Sci. 2017 Feb 15;488:48-60. Epub 2016 Oct 28. https://doi.org/10.1016/j.jcis.2016.10.082 ; PubMed PMID: 27821339 , Feb-2017
Articles in Peer-reviewed Journals	Kundan A, Nguyen TT, Plawsky JL, Wayner PC, Chao DF, Sicker RJ. "Condensation on highly superheated surfaces: unstable thin films in a wickless heat pipe." Phys Rev Lett. 2017 Mar 3;118(9):094501. https://doi.org/10.1103/PhysRevLett.118.094501 ; PubMed PMID: 28306276 , Mar-2017